

his question paper contains 4 printed pages.]

Your Roll No.....

No. of Question Paper : 6624

HC

Unique Paper Code : 32351302

Name of the Paper : Group Theory 1

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

Write your Roll No. on the top immediately on receipt of this question paper.

Attempt any **two** parts from each question.

All questions are compulsory.

(a) For a fixed point (a, b) in $\mathbb{R}^2$, define $T_{(a,b)} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $(x, y) \rightarrow (x + a, y + b)$.

Show that $T(\mathbb{R}^2) = \{T_{a,b} | a, b \in \mathbb{R}\}$

is a group under function composition. (6)

(b) (i) Find the inverse of $\begin{pmatrix} 2 & 6 \\ 3 & 5 \end{pmatrix}$ in $GL(2, \mathbb{Z}_{11})$. (4)

P.T.O.

- (ii) Let G be an Abelian group under multiplication with identity e . Show that

$$H = \{x^2 \mid x \in G\} \text{ is a subgroup of } G.$$

- (c) (i) Let G be a group. Show that $Z(G) = \bigcap_{a \in G} C(a)$

where $Z(G)$ is the Center of G and $C(a)$ is the Centralizer of a .

- (ii) Let G be the group of nonzero real numbers under multiplication. Show that

$$H = \{x \in G \mid x = 1 \text{ or } x \text{ is irrational}\}$$

and $K = \{x \in G \mid x \geq 1\}$ are not subgroups of G .

2. (a) Let G be a group and let $a \in G$. If $|a| = n$, prove that

$$\langle a \rangle = \{e, a, a^2, \dots, a^{n-1}\} \text{ and } a^i = a^j \text{ if and only if } n \text{ divides } i - j.$$

- (b) Suppose that $|a| = 24$. Find a generator for $\langle a^{21} \rangle \cap \langle a^{10} \rangle$.

- (c) If $|a| = n$, show that

$$\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$$

and that

$$|a^k| = \frac{n}{\gcd(n,k)}$$

3. (a) Define the Alternating Group A_n . Show that it forms a subgroup of the Permutation Group S_n and $|A_n| = \frac{n!}{2}$. (6)
- (b) Prove that every group is isomorphic to a group of permutations. (6)
- (c) Prove that $U(10)$ is not isomorphic to $U(12)$. (6)
4. (a) State and prove Orbit Stabilizer Theorem. (6½)
- (b) (i) Prove that $aH = H$ if and only if $a \in H$. (3)
- (ii) Prove that $aH = bH$ or $aH \cap bH = \phi$. (3½)
- (c) (i) Prove that order of $U(n)$ is even when $n > 2$. (3)
- (ii) Prove that a group of prime order is cyclic. (3½)
5. (a) Let H and K be subgroups of a finite group G and let
- $$HK = \{hk \mid h \in H, k \in K\}$$
- and
- $$KH = \{kh \mid k \in K, h \in H\}.$$
- Prove that HK is a group if and only if $HK = KH$. (6½)

(b) Let ϕ be a homomorphism from a group G to a group $\bar{G}$ and let g be an element of G . Prove that

(i) If $\phi(g) = g'$, then $\phi^{-1}(g') = \{x \in G \mid \phi(x) = g'\} = g\text{Ker}\phi$ (4)

(ii) If $|\text{Ker}\phi| = n$, then ϕ is an n -to-1 mapping from G onto $\phi(G)$. (2½)

(c) (i) Prove that A_n is normal in S_n . (3½)

(ii) If G is a non-Abelian group of order p^3 (p is prime) and $Z(G) \neq \{e\}$, prove that $|Z(G)| = p$. (3)

6. (a) State and prove The First Isomorphism Theorem. (6½)

(b) Let G be a group and let $Z(G)$ be the center of G .

Prove that if $G/Z(G)$ is cyclic, then G is Abelian. (6½)

(c) Let $4\mathbb{Z} = \{0, \pm 4, \pm 8, \dots\}$. Find $\mathbb{Z}/4\mathbb{Z}$. (6½)